Newumal Weerasinghe

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PROFESSIONAL SUMMARY

Associate Tech Lead and Team Lead with 7+ years building and leading SaaS products, and an early hands-on AI adopter. For the last 2+ years I've applied AI across design, product thinking, and delivery while keeping the real decisions human-owned. Most of my work has been on AI features (LLMs, RAG, agents, Copilot Context), cost optimization, and performance fixes, all on cloud-native AWS with a React.js, and Node.js stack across PostgreSQL and NoSQL (MongoDB, DynamoDB). Comfortable working between product and engineering: turning business goals into things that actually ship, and bringing the team along while doing it.

WORK EXPERIENCE

Associate Tech Lead & Team Lead, Jan 2026 – Present
Velaris | Moratuwa, Sri Lanka

- Lead a cross-functional engineering team, owning sprint planning, technical direction, code reviews, and mentoring.
- Pitched a new AI feature and took it from a proof-of-concept the CEO, Head of AI, and product owners backed, through user feedback and fine-tuning, to production. Worked with the AI team to reuse the approach across other teams, and it now helps with new-customer onboarding and sales.
- Set up a data-driven way to prioritize churn and reliability work. Went through the priority-issues channel, wrote up root causes with proper fixes, effort estimates for dev and QA, and timelines, then got the product owner and engineering lead aligned to act on them. That moved us off patching symptoms and toward real fixes, which freed up capacity for new features. Other teams later adopted it too.
- Ran several stability, cost, and scalability efforts with a product and cost mindset, including customer 360-view improvements, the v4 platform rollout, and scaling to support 500K+ records.
- Mentored engineers across a few teams on integration work, email/calendar/transcripts, and testing strategy, and helped with hiring by running engineering interviews.
- Pushed for responsible AI use across the team. At the engineering lead's and product owner's request, ran sessions on using AI throughout the engineering lifecycle without giving up engineering judgment, set up a best-practices framework and standards, and built custom agents tuned for accurate, low-hallucination output. Most of the team now uses AI well day to day.
- Delivered my own features on time and defect-free across 2+ years of AI-assisted work, leaning on AI for design, brainstorming, weighing trade-offs, and the repetitive parts, while keeping every engineering decision my own.

Senior Software Engineer (Full Stack) Jul 2022 – Jan 2026
Velaris | Moratuwa, Sri Lanka

- Built an AI Copilot feature that turns natural-language questions into charts (line, pie, bar, KPI) and interactive Canvas dashboards, and improved the Copilot's context so users can ask more detailed questions.
- Filled in as acting team lead while a senior lead was out: ran the Scrum ceremonies, split work across the engineers, coordinated with QA and PM, and shipped several production releases under pressure without breaking stability.
- Built user-facing AI features hands-on across multiple LLMs, RAG, context graphs, and agentic systems, with a lot of attention on keeping LLM costs down.
- Led user-integration projects end to end on Google Cloud and Microsoft Azure, building and verifying OAuth apps (app registration, consent screens, Google/Microsoft app-verification review, and auth/token flows) for Gmail, Outlook, calendars, and meeting transcripts.
- Delivered AI meeting-intelligence integrations for AI and Microsoft Teams transcripts, plus Google and Outlook calendars and Gmail, using new microservice pipelines and frontend work. The Teams integration went out on a tight deadline with no post-release issues.

- Designed and shipped a Salesforce integration suite, delivering 5 components against 3 planned (Notes sync and pushback, Opportunity and Risk sync and pushback), including a streaming-data design that handled large datasets with very few defects.
- Reworked core email processing, including a global event-handler fix that drops unwanted events during the initial sync. It brought real performance and cost gains, with no bugs reported after release.
- Led AWS cost-optimization work: right-sized resources and cleaned up logging to cut operational costs across compute, I/O, and database layers. Also built a token-health monitoring system that catches revocations early and alerts before they cause problems.
- Created a reusable Git integration template that standardized the Teams, Email, and Calendar patterns, so the team could build user-level integrations faster and more consistently.
- Built SaaS features with React.js, Node.js (NestJS / Express), Redux, PostgreSQL, and AWS (Lambda, EC2, S3, SQS, Step Functions, CloudWatch, CloudFormation), keeping test coverage high with Jest and working closely with QA.

Software Engineer (Full Stack) Mar 2021 – Jul 2022

Sezenta (Pvt.) Ltd | Battaramulla, Sri Lanka

- Developed web applications with React.js, Node.js, MongoDB, and AWS.
- Built cross-platform mobile apps with React Native and Node.js.
- Wrote REST APIs with Node.js, Express, and NestJS.
- Used AWS services to improve app scalability and performance.

Development Engineer & Co-Founder Feb 2019 – Mar 2021

STEM Labs (Pvt.) Ltd | Nugegoda, Sri Lanka

- Delivered IT and STEM education lectures covering robotics and electronics.
- Led web design and development projects using PHP, JavaScript, CSS, and WordPress.
- Developed software and hardware solutions, including a patented IoT hardware product.

Associate Software Engineer Feb 2017 – Feb 2019

Effective Solutions (Pvt.) Ltd | Moratuwa, Sri Lanka

- Built software using the Symfony Framework, including a loyalty platform and student enrollment system for Trinity College.
- Developed e-commerce websites for clients including Softlogic and TyreOnline.
- Designed IoT platforms using Arduino, MQTT, Node.js, and Node-RED; integrated GPS tracking via REST API.

TECHNICAL SKILLS

Languages & Frameworks: JavaScript (ES6+), React.js (Hooks, Context API, custom hooks, reusable components), Redux, Node.js, Express.js, NestJS, React Native, PHP, Python (learning)

Databases: PostgreSQL, MongoDB, DynamoDB, MySQL, TypeORM

Cloud & DevOps: AWS (Lambda, EC2, S3, SQS, Step Functions, CloudWatch, CloudFormation), Google Cloud Platform, Microsoft Azure, CI/CD, Docker

Identity & Auth: OAuth 2.0, JWT, Google & Azure app registration and verification, consent screens, SSO, token management , role based user management

AI Engineering: LLM integration, RAG, context graphs, custom AI agents, prompt engineering, evals, LLM cost management, AI-assisted SDLC (GitHub Copilot, Cursor, Claude Code)

Integrations & APIs: REST APIs, third-party integrations (Google, Outlook, Teams Transcript APIs, Salesforce)

Testing & Tools: Jest (unit & integration testing), ESLint, Prettier, Git, JIRA, Confluence

Leadership: Technical leadership, team management, agile/Scrum delivery, system design, clean code (SOLID principles), code review, mentoring

EDUCATION

BSc in Electronics & IT

University of Colombo | Colombo, Sri Lanka

ACHIEVEMENTS

Technical content creator: Publish engineering and technical-leadership content on YouTube (@buildwithnewumal).

Kotiyak Watina Adahasak, Season 2 Final 24 (2018): Reached the final 24 of a national innovative-ideas competition broadcast on Sirasa TV, Sri Lanka.